MACHINE LEARNING QUESTION BANK (UNIT IV, V, III (Half))

PART – A

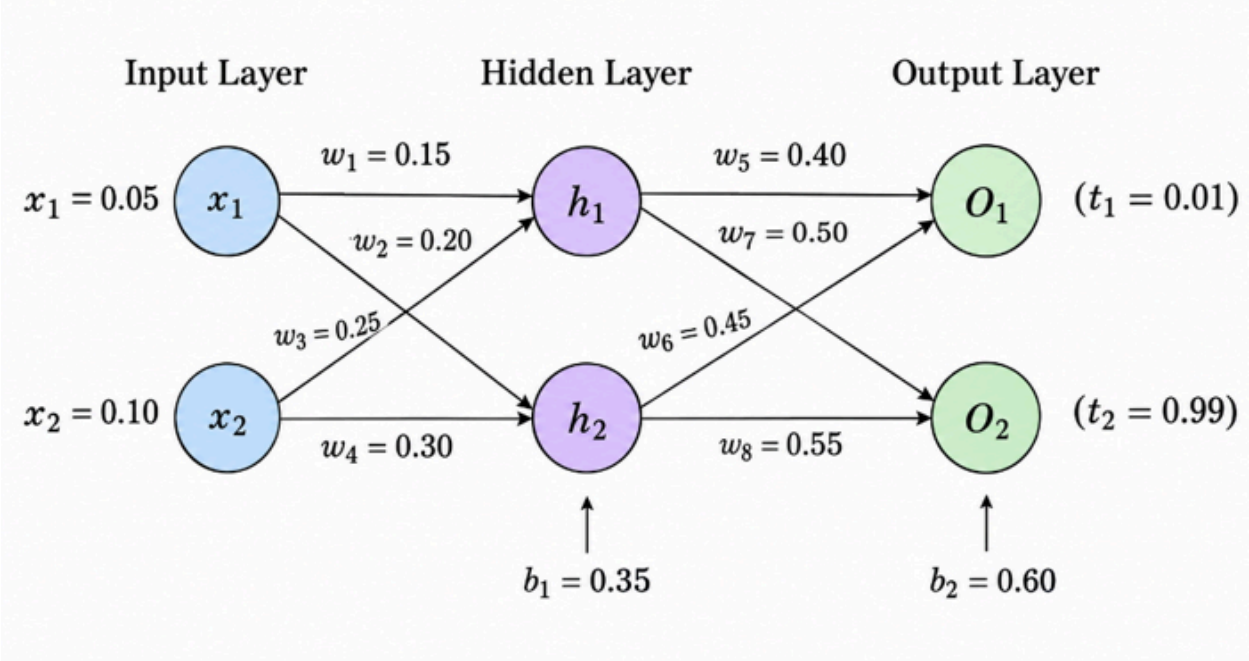
1.	A neuron receives inputs $x_1 = 2$, $x_2 = 3$, weights $w_1 = 0.4$, $w_2 = 0.5$, and bias $b = 0.2$. Using the sigmoid activation function, calculate the neuron output.
2.	A hidden neuron receives a weighted sum of $z = -2.5$. Calculate its output using sigmoid and ReLU activation functions and state the difference.
3.	A binary-classification model predicts $\hat{y} = 0.8$ for an actual class $y = 1$. Calculate the binary cross-entropy loss.
4.	A neural network has a learning rate of 0.01 and a gradient of 0.4 for a particular weight. Calculate the new weight if the current weight is 0.6.
5.	A dataset contains 100 samples. If 5-fold cross-validation is performed, determine the number of training and validation samples in each fold.
6.	A grid search evaluates the following learning rates and maximum depths: Learning rate: 0.01, 0.1, 0.2 Maximum depth: 3, 5, 7 If the validation accuracies are given as [85%, 88%, 87%, 89%, 92%, 90%, 86%, 91%, 89%], identify the parameter combination with the highest validation accuracy.
7.	In AdaBoost, a training sample has a weight of 0.10 and is misclassified by the current weak learner. Explain how its weight is adjusted in the next iteration.
8.	An RL agent has two actions: A_1 gives a known reward of 5 and A_2 is unexplored. If the agent chooses A_2 to obtain new information, identify the type of strategy being used.
9.	A two-dimensional dataset has eigenvalues 8 and 2. Determine the percentage of total variance retained when only the first principal component is selected.
10.	How does density-based clustering identify noise or outlier data points?

11.	Consider a neuron with $x_1 = 1$, $x_2 = 2$, weights $w_1 = 0.5$, $w_2 = 0.4$, and bias $b = 0.1$. Calculate the output using ReLU activation.
12.	For a simple model with $z = wx$, where $x = 2$, $w = 0.5$, and loss $L = (z - 1)^2$, calculate $\partial L/\partial w$ using the chain rule.
13.	A neural network has a weight value of 0.8 and uses L2 regularization with $\lambda = 0.01$. Calculate the regularization contribution λw^2 to the loss.

<table>
<tr><td>14.</td><td>A neural network produces the following accuracies:
<table>
<tr><th>Epoch</th><th>Training Accuracy</th><th>Validation Accuracy</th></tr>
<tr><td>1</td><td>75%</td><td>73%</td></tr>
<tr><td>10</td><td>90%</td><td>86%</td></tr>
<tr><td>20</td><td>96%</td><td>80%</td></tr>
<tr><td>30</td><td>99%</td><td>72%</td></tr>
</table>
Identify the likely training problem and justify your answer.</td></tr>
<tr><td>15.</td><td>A dataset contains 80 samples from Class A and 20 samples from Class B. Explain why stratified k-fold cross-validation is more suitable than ordinary k-fold cross-validation.</td></tr>
<tr><td>16.</td><td>A model has 5 hyperparameters, each with 4 possible values. Determine the total number of combinations evaluated by exhaustive grid search.</td></tr>
<tr><td>17.</td><td>Three decision trees produce predictions with high variability on different training samples. Explain how Bagging can improve the overall prediction.</td></tr>
<tr><td>18.</td><td>Given an immediate reward $R = 5$, discount factor $\gamma = 0.8$, and next-state value $V(s') = 10$, calculate the value of the current state using $V(s)=R+\gamma V(s')$</td></tr>
<tr><td>19.</td><td>The eigenvalues obtained from PCA are 8, 4, 2, and 1. Calculate the percentage of total variance explained by the first principal component.</td></tr>
<tr><td>20.</td><td>A medical test has a disease prior probability of 0.10, sensitivity of 0.90, and false-positive rate of 0.05. Calculate the probability that a person has the disease given that the test result is positive.</td></tr>
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PART – B

1.A neural network has two input neurons, two hidden neurons, and two output neurons as shown in the given figure. For inputs $x_1 = 0.05$ and $x_2 = 0.10$, target outputs $t_1 = 0.01$ and $t_2 = 0.99$, and the given weights and biases, perform a complete forward propagation and calculate the MSE. Then apply backpropagation to determine the weight updates for one iteration using the sigmoid activation function and a specified learning rate.



2. A neural network produces the following results:

Epoch	Training Accuracy	Validation Accuracy
1	70%	69%
10	85%	82%
20	93%	84%
30	97%	78%
40	99%	71%

The network uses 3 hidden layers, 128 neurons per layer, a learning rate of 0.1, and no regularization.

Identify the problem shown by the model. Propose suitable changes to the learning rate, network size, dropout, L2 regularization, and early stopping. Explain how each change is expected to affect model generalization.

3. A binary classification dataset contains 1,000 samples, consisting of 700 samples from Class 0 and 300 samples from Class 1.
Perform a 5-fold cross-validation and explain the distribution of training and validation samples in each fold. Then explain how stratified 5-fold cross-validation would maintain the class proportions. Discuss the advantages and limitations of both methods when estimating model performance.

4. Consider an agent navigating through a sequence of states $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4$ (Goal). At each state, the agent can choose either Left or Right. The agent receives a reward of +10 upon reaching the goal state S_4 and a reward of −1 for every other movement. The discount factor is $\gamma = 0.9$. Formulate the given problem as a Markov Decision Process by identifying the states, actions, rewards, and transition probabilities. Explain how the agent balances exploration and exploitation while navigating toward the goal. Using the Bellman equation, calculate the value of a state when the immediate reward is −1 and the value of the next state is 8.

5. Apply Principal Component Analysis (PCA) to this dataset and determine:

	EX1	EX2	EX3	EX4
X_1	4	8	13	7
X_2	11	4	5	14

1. The mean of each feature.
2. The centered data matrix.
3. The covariance matrix.
4. The eigenvalues and eigenvectors.
5. The principal components.

6. The percentage of variance explained by each component.

The reduced one-dimensional representation of the data.

6. Consider an HMM with two hidden states:

S_1 = Rainy

S_2 = Sunny

Initial probabilities:

$P(S_1) = 0.6,\ P(S_2) = 0.4$

Transition probabilities:

$P(S_1 \mid S_1) = 0.7,\ P(S_2 \mid S_1) = 0.3$

$P(S_1 \mid S_2) = 0.4,\ P(S_2 \mid S_2) = 0.6$

Emission probabilities:

State	Walk	Shop
S_1	0.2	0.8
S_2	0.7	0.3

For the observation sequence:

$Walk \rightarrow Shop$

Apply the Forward Algorithm and calculate the probability of observing the sequence $Walk \rightarrow Shop$. Show all intermediate calculations.

7. Consider a feed-forward neural network having two input neurons, two hidden neurons, and two output neurons. The network uses the sigmoid activation function at both the hidden and output layers.

The input values are:

$x_1 = 0.05,\ x_2 = 0.10$

The weights connecting the input layer to the hidden layer are:

- Weight from x_1 to h_1: $w_1 = 0.15$
- Weight from x_2 to h_1: $w_2 = 0.20$
- Weight from x_1 to h_2: $w_3 = 0.25$

- Weight from x_2 to h_2: $w_4 = 0.30$

The bias for the hidden layer is:

$b_1 = 0.35$

The weights connecting the hidden layer to the output layer are:

- Weight from h_1 to O_1: $w_5 = 0.40$
- Weight from h_2 to O_1: $w_6 = 0.45$
- Weight from h_1 to O_2: $w_7 = 0.50$
- Weight from h_2 to O_2: $w_8 = 0.55$

The bias for the output layer is:

$b_2 = 0.60$

The desired target outputs are:

$t_1 = 0.01,\ t_2 = 0.99$

The sigmoid activation function is given by:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Perform the following:

1. Calculate the net input and output of hidden neurons h_1 and h_2.
2. Calculate the net input and predicted output of output neurons O_1 and O_2.
3. Calculate the error using the Mean Squared Error (MSE).
4. Perform one complete backpropagation step.
5. Calculate the gradients for the output-layer weights.
6. Calculate the gradients for the hidden-layer weights.

Update all the weights using a learning rate of $\eta = 0.5$.

8. Consider a binary-classification problem with 20 input features and two output classes.

Design a neural-network architecture containing:

- An input layer
- Two hidden layers
- An output layer

Select suitable activation functions from sigmoid, tanh, and ReLU for the hidden and output layers. Explain the mathematical operation of each selected activation function, discuss their advantages and limitations, and justify the architecture and activation-function choices.

9. A Random Forest model has the following hyperparameters:

- Number of trees: 100, 200, 300
- Maximum depth: 5, 10, 15
- Minimum samples split: 2, 5

The validation accuracies obtained for selected configurations are:

Trees	Depth	Min. Samples Split	Accuracy
100	5	2	88%
100	10	5	91%
200	10	2	93%
200	15	5	92%
300	10	5	94%
300	15	2	93%

Determine the configuration selected based on validation accuracy. Calculate the total number of combinations possible in the complete grid and explain how random search can reduce computational cost.

10. A binary-classification dataset contains 10,000 samples, 30 input features, and a class distribution of 70% Class 0 and 30% Class 1. A single decision tree gives 82% accuracy but shows high variation across different training sets.
Explain how Bagging, AdaBoost, and XGBoost would approach this classification problem. Compare their ensemble-building mechanisms and error-reduction strategies. Discuss the factors that should be considered when selecting an ensemble method for this dataset.

11. Consider the following two-dimensional dataset:

Sample	X_1	X_2
1	2	1

2	3	2
3	4	3
4	5	4

The covariance matrix obtained after preprocessing is:

$$C = \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix}$$

Apply PCA to the given dataset and:

a) Calculate the eigenvalues of the covariance matrix.

b) Calculate the corresponding eigenvectors.

c) Determine the principal component that captures the maximum variance.

d) Calculate the percentage of variance explained by the first principal component.

e) Project the given data onto the first principal component.

12. A spam-filtering system classifies emails as Spam (S) or Not Spam (N) based on whether an email contains the word "Offer" (O).

From previous data:

$P(S) = 0.30$

$P(N) = 0.70$

The probability that the word "Offer" appears in a spam email is:

$P(O \mid S) = 0.80$

The probability that the word "Offer" appears in a non-spam email is:

$P(O \mid N) = 0.10$

An incoming email contains the word "Offer".

Using Bayesian inference:

1. Calculate the probability that the email is spam, $P(S \mid O)$.
2. Calculate the probability that the email is not spam, $P(N \mid O)$.
3. Determine the classification of the email based on the higher posterior probability.

Explain how the prior probability and likelihood influence the classification.